

Horizon: Evidence-Conditioned Runtime Assurance for Autonomous Maritime Systems

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ABSTRACT

Horizon inserts an independent safety governor between a vessel's decision-making AI and its actuators. It joins navigation, contact, platform, communication, perception, and neural-health evidence before passing, modifying, replacing, or rejecting each proposed command. An initial 60-branch, idealized-timing development study compared five candidates across four encounter classes. No declared-recoverable branch collided and no unsafe or stale decision passed the independent gate; A5, an evidence-conditioned predictive and barrier-filter hybrid, recorded 99.83% gate acceptance with three deadline misses. A later paired study with nonzero front-end service delay found that A5 intervened earliest in the in-domain crossing case, 39.7 s before a sampled recovery boundary, but also missed seven candidate deadlines. H5 now supplies a temporally trained, causally tested neural-health shadow, while A6 adds a non-actuating Singapore policy shadow around the A5 result. A5 remains the leading development candidate, not a selected or certified architecture; the frozen held-out comparison remains blocked on calibration.

CCS CONCEPTS

• **Computer systems organization** → **Robotic autonomy**; • **Computing methodologies** → *Computer vision*.

KEYWORDS

runtime assurance, autonomous maritime systems, safety governor, mechanistic interpretability, sparse autoencoder

1 SAFETY OBJECTIVE

Autonomous maritime systems can emit unsafe commands even when individual components appear healthy. Horizon therefore enforces one protected actuation path. The external AI proposes a command but cannot write to steering or propulsion. Assurance evaluates public observations and declared uncertainty; an exclusive gate revalidates identity, lineage, epoch, freshness, deadline, actuator limits, and final-command safety. A watchdog retains minimum-risk authority if supervision stops.

The governing rule is monotone: unknown, stale, invalid, unbounded, or late evidence may only preserve or reduce authority. Camera or neural evidence is a *perception-health input*, not proof of obstacle-free water.

2 CANDIDATE ARCHITECTURES

The five actuation candidates implement one bounded input-to-decision contract. A1 applies CPA/TCPA thresholds and Simplex switching. A2 uses an uncalibrated analytic Gaussian collision-risk

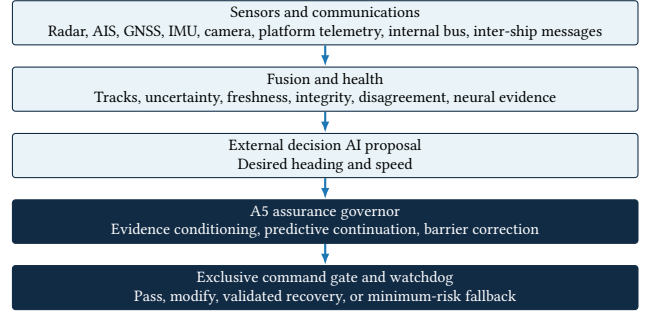


Figure 1: Protected Horizon control path. Evaluation truth is isolated from this path.

approximation with Simplex recovery. A3 requires finite rollout and a complete recovery continuation. A4 searches a nonlinear vessel plant map for a barrier-compatible command and validates it with a full 3DOF rollout. A5 conditions bounds and speed on evidence health, attempts predictive recoverability, and invokes A4 when required. A6 is deliberately different: it observes an identity-matched A5 result and emits a policy-shadow assessment with no command or control authority.

2.1 A5 decision path

Healthy evidence retains nominal bounds and a 6 m/s cap; degraded evidence expands bounds by 1.5 and caps speed at 3 m/s; invalid evidence caps speed at 1 m/s and forces recovery. A5 checks the request through a short handoff and complete recovery continuation. If that fails, it applies A4 and validates the correction with a complete rollout. Incomplete, infeasible, late, or unsupported results fall back to recovery or minimum-risk control.

3 EXPERIMENTAL METHOD

The matrix contained 12 paired identities: four scenarios, three seeds, and five candidates per identity, for 60 branches. Scenarios covered recoverable crossing, dense traffic, vessel degradation, and initially-unrecoverable encounters. Each branch ran for 30 simulated seconds under an idealized profile with zero sensing, fusion, AI, recovery-prime, and actuator service time, plus 20 ms candidate and gate service. Exact joins linked evaluation, diagnostics, and assumption audits; all collisions, timeouts, bound violations, and censored outcomes remained in their denominators [1]. These results are retained as an architecture-development reference, not as evidence for end-to-end timing.

4 RESULTS AND DEVELOPMENT STATUS

All 15 branches in the deliberately unrecoverable stratum collided; none of the 45 declared-recoverable branches collided. No unsafe or stale result was accepted by the gate. A1 and A2 were computationally light but mostly passed proposals that the independent gate later rejected, giving acceptance rates below 9%. A3–A5 actively selected recovery or minimum-risk actions and exceeded 99.4% acceptance. A5 recorded the highest acceptance and fewer deadline misses than A3 or A4 (Fig. 2). We therefore continue A5 in prototype integration while retaining A1 as the transparent baseline; the later evidence below governs any selection claim.

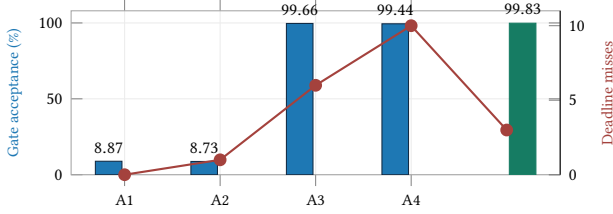


Figure 2: Idealized-profile gate acceptance (bars) and candidate deadline misses (red markers), 12 branches per candidate. Counts are descriptive, not a formal safety ranking.

4.1 Nonzero-delay development update

A later paired study retained 20 ms sensing and fusion service, 40 ms AI service, and 20 ms candidate service before synchronous local recovery, gate, and dispatch calls [3]. The profile is an acceptance load for single-process integration, not a production latency measurement. In the in-domain crossing case, A5 produced the earliest protected intervention and the largest accepted-command count (Table 2). A3 and A4 also produced accepted recovery commands. A1 and A2 repeatedly classified the straight proposal as clear, but the independent predictive gate rejected it.

All five paired dense-traffic branches were outside the declared engineering bounds and accepted no candidate decision. Across both scenarios, all ten branches avoided collision, boundary, and grounding violations, but every route remained incomplete at its horizon. These outcomes preserve A5 as the development lead while withholding a scientific winner. The frozen comparison now specifies 1,200 episode identities and 6,000 candidate branches, at least 30 paired seeds per headline cell, strict acceptance gates, and a Pareto comparison only among survivors; it cannot run until controller-input and perception-health calibration artifacts are frozen [4].

A separate 25-branch Singapore full-pipeline smoke exercised five evidence conditions—AIS absent, stale/dropout, contradictory, overloaded synthetic traffic, and radar-only—across A1–A5.

Table 1: A1–A5 development results.

ID	Gate accepted	Deadline misses	Recoverable collisions	Unsafe/stale accepted
A1	8.87%	0	0	0
A2	8.73%	1	0	0
A3	99.66%	6	0	0
A4	99.44%	10	0	0
A5	99.83%	3	0	0

Table 2: Crossing results under the local acceptance-load profile.

ID	Accepted	Deadline misses	First intervention / lead time
A1	0	0	none / unknown
A2	0	2	none / unknown
A3	36	10	18.7 s / 21.3 s
A4	27	3	20.7 s / 19.3 s
A5	183	7	0.3 s / 39.7 s

Table 3: H1–H5 development experiments and present interpretation.

ID	Experiment	Result
H1	Conventional exposure, blur, freeze, timing, horizon, occlusion, and output-change checks.	0/296 complete
H2	Diagonal Mahalanobis distance on a grouped encoder summary.	median 7.233
H3	PCA reconstruction of the same summary.	median 0.01233
H4	Encoder-layer SAE with matched causal controls.	15/18 wins
H5	Temporal-fusion TopK SAE, cross-seed stability, and matched causal controls.	18/18 wins

It loaded and truth-scored every branch with zero recorded collision, grounding, or boundary events, but all missions were censored and no candidate decision reached the gate because the then-current fusion snapshot-origin check failed closed. The run validates fixture and pipeline execution only; it is not controller-performance or held-out evidence. A 25.9 s AISStream connectivity smoke observed three valid positions and retained neither raw vessel records nor the API key; it establishes envelope-shape feasibility, not navigation evidence.

5 NEURAL MECHANISTIC EVIDENCE

Horizon evaluates perception health as a sequence of increasingly mechanistic tests. All reported numbers use the frozen kope81 development collection and are not held-out safety claims.

H1 remained incomplete because independent horizon and occlusion evidence was unavailable. H2 and H3 detected representation shift but supplied neither a semantic mechanism nor a calibrated fault threshold. The validated 64-feature H4 direction beat both random-direction and equal-norm controls in 15 of 18 trials ($p = 0.00377$); a tuned 96-feature model reconstructed 6.1% better but failed its causal gate at 11 of 18 ($p = 0.240$). Reconstruction quality alone was therefore insufficient.

5.1 H5 temporal-feature monitor

H5 moves the intervention point from the image encoder to WaSR-T’s temporal_fusion module. Each frame yields a spatially averaged 2,048-channel state. A 128-latent SAE applies hard TopK sparsity ($k = 16$) and optimizes reconstruction together with a temporal triplet loss: adjacent frames are pulled together while the same sampled position in another development sequence is pushed apart. This follows evidence that video SAEs require explicit temporal coherence and that useful internal features should be validated by intervention rather than reconstruction alone [8–10].

The selected H5 configuration converged in 622 epochs with 3/128 dead features. Against a capacity-matched zero-temporal-weight baseline, validation reconstruction MSE improved from 0.9533 to 0.8295 and the cross-sequence to adjacent-code distance

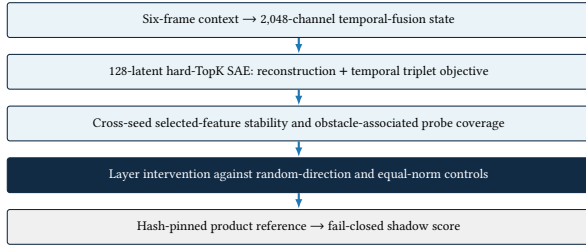


Figure 3: H5 evidence path. No stage grants navigation or actuation authority.

ratio improved from 6.12 to 6.89. Whole-dictionary seed stability remained poor (mean nearest decoder cosine 0.425), so H5 gates the specific selected feature. Feature 55 matched both alternate seeds with minimum decoder cosine 0.528 and activation correlation 0.667, then beat both causal controls in all 18 trials ($p = 3.81 \times 10^{-6}$). Two earlier development runs informed this feature-specific protocol; the successful rerun is consequently development evidence, not a sealed confirmation.

H5 is now wired into the product launcher, shared health contract, monitor registry, and recorded-camera service using a hash-pinned reference artifact. It runs *shadow-only*: the first frame is unknown until temporal context exists, and every later score remains unknown without frozen calibration. Before authority can be considered, H5 still needs fault-labelled calibration, independent H1–H5 held-out comparison, target-computer latency qualification, and spatially resolved causal/collateral-effect tests. A5 therefore receives no authority upgrade from H5.

6 A6 SINGAPORE POLICY SHADOW

A6 is an implemented, deterministic observation layer around A5, not a sixth actuation candidate. Its bounded first slice models a generic 12 m Singapore government/non-commercial, power-driven USV under remote supervision in daylight and clear visibility. This engineering profile matches Horizon’s simulated vessel; it is not a claim that the named Republic of Singapore Navy MARSEC USV has those dimensions. Public information instead describes that operational platform as 16.9 m, 30 tonnes, remotely supervised, and equipped with rules-based collision avoidance intended to follow COLREG behaviour [11].

A6 consumes the original governor identity, a completed A5 decision, explicit operational-profile facts, rule evidence, an evaluation time, and a hash-pinned policy bundle. It produces only an A6PolicyShadowAssessment; it is not registered in the candidate or gate allowlists and contains no issued command. If A5 rejects, recovers, becomes invalid, or does not match the input identity, A6 withholds support. Thus a policy result cannot weaken A5’s physical-safety decision.

The first rule set implements three-valued engineering proxies for lookout evidence availability (COLREG 5), safe speed (6), collision-risk doubt (7), early and substantial action (8), and clear-visibility encounter classification and duties (13–17). Applicability is *applicable*, *not applicable*, or *unknown*; ambiguous encounters, missing or stale evidence, invalid source hashes, and restricted visibility return unknown and withhold support. These findings are

not legal determinations and must not be described as COLREG compliance.

The source hierarchy separates runtime authority from design references. Singapore Port Regulations specifically apply regulations 27, 28, 29, 43, 49, and 49A to Singapore Armed Forces vessels [12]; future local-rule ingestion must preserve that applicability rather than treating every port rule as binding. The 2026 non-mandatory IMO MASS Code is a design-assurance reference only [13]; it is not runtime law or command ground truth. Exact source texts and bundle content require reviewed SHA-256 pins before use. Weapons, targeting, classified doctrine, and rules of engagement are excluded.

7 LIMITATIONS AND NEXT STUDY

The experiments cannot support certification or a final ranking. All 60 initial missions and all ten nonzero-delay paired missions were censored, so route completion and mission cost are unknown. Forty initial branches exceeded an engineering uncertainty bound, and the later dense-traffic case remained out of domain. The initial study used idealized front-end timing; the later acceptance load still models local recovery, gate, and dispatch as synchronous calls. Gate/watchdog actions affect trajectories, and the sampled finite-library recovery boundary is not a viability-kernel proof. Three seeds per initial scenario and one seed per later scenario are insufficient for statistical claims.

The next study must complete controller-input and perception calibration before opening held-out data; run the frozen 6,000-branch comparison with at least 30 seeds per headline cell; resolve route censoring; separate preventable and unrecoverable encounters; validate matched-false-positive-rate perception thresholds on target CUDA hardware; and measure sensor-to-actuator latency on a reserved timing host. A6 additionally requires reviewed source snapshots and applicability, upstream policy-context contracts, authoritative TSS/channel geometry, deterministic golden legal-review cases, and a resource-isolated A5-versus-A6 shadow study. Generic two-vCPU CI verifies fail-closed configuration and lifecycle behavior but does not establish the 40 ms end-to-end timing claim [7]. Deployment also requires qualified hydrodynamics, uncertainty coverage, communication-fault tests, and real ship interfaces.

8 CONCLUSION

Horizon demonstrates a coherent pattern: diverse ship and AI evidence informs a bounded governor while an independent gate owns actuation and fails closed. A5 remains the leading development architecture because it combines evidence-conditioned margins, predictive continuation, plant-aware correction, and the earliest measured intervention in the nonzero-delay crossing case. H5 and A6 now add independent neural-health and policy shadows without expanding actuation authority. The updated evidence also makes the boundary clearer: A5 missed deadlines, no candidate produced accepted decisions in the out-of-domain dense case, every evaluated mission was censored, and the frozen held-out comparison remains blocked on calibration. Neither a causal feature nor a policy proxy is yet an operational or legal assurance claim.

9 IMPLEMENTATION SNAPSHOT

The product H5 path was exercised with the pinned WaSR-T checkpoint and H5 reference on eight recorded inputs. Three frames completed before five bounded drop-oldest queue evictions, producing six accepted observation records with no inference, publication, expiry, or staleness failures. The first health record was unknown because no previous embedding existed; two later records emitted reconstruction and temporal-distance statistics and remained unknown because calibration is intentionally absent. This is a functional live-shadow check, not throughput qualification.

The A6 shadow implementation adds 16 focused tests; 21 focused selections and all 154 assurance tests pass. Tests cover deterministic replay, three-valued applicability, invalid or stale bundles, excluded domains, encounter ambiguity, and the invariant that A6 cannot support a result rejected by A5. A6 remains outside the live candidate registry pending shared policy-context contracts, reviewed source ingestion, and shadow evaluation.

REFERENCES

- [1] Horizon Project, "A32 A1–A5 30-second development analysis," revision 28eb923, Sept. 2026, analysis hash 8004358e...e909.
- [2] Horizon Project, "MODD2 development evidence and extraction completion," Sept. 2026.
- [3] Horizon Project, "R1/R2 local acceptance-load result," Sept. 22, 2026, summary hash 73766b98...c684.
- [4] Horizon Project, "R4/R6 execution readiness," Sept. 22, 2026.
- [5] Horizon Project, "MODD2 development extraction completion," Sept. 22, 2026, report hash 1197d33e...0220.
- [6] Horizon Project, "Perception calibration acquisition progress," Sept. 22, 2026, plan hash cf68b0bf...744.
- [7] Horizon Project, "Hosted Linux system-test stability and timing boundary," Sept. 22, 2026.
- [8] E. Dokme and S. Vishwanath, "Spatio-Temporal Sparse Autoencoders for Video," arXiv:2604.03919, 2026.
- [9] S. Marks et al., "Sparse Feature Circuits," arXiv:2403.19647, 2024.
- [10] S. Heimersheim and N. Nanda, "How to Use and Interpret Activation Patching," arXiv:2404.15255, 2024.
- [11] Singapore Ministry of Defence, "The Republic of Singapore Navy's Unmanned Surface Vessels Progressively Operationalised to Enhance Maritime Security," Feb. 4, 2025, https://www.mindef.gov.sg/news-and-events/latest-releases/04feb25_fs/.
- [12] Singapore Statutes Online, "Maritime and Port Authority of Singapore (Port) Regulations," current version accessed Sept. 24, 2026, <https://sso.agc.gov.sg/SL/MPASA1996-RG7>.
- [13] International Maritime Organization, "FAQ—Autonomous shipping," MASS Code resolution MSC.595(111), effective July 1, 2026, <https://www.imo.org/en/mediacentre/hottopics/pages/autonomous-shipping.aspx>.
- [14] L. Zust et al., "WaSR-T official implementation," <https://github.com/lojzezust/WaSR-T>.
- [15] B. Bowcon et al., "MODD2 maritime obstacle-detection dataset," <https://github.com/bborja/modd>.